

SOCIAL STUDIES RESEARCH

EL NIÑO DROUGHT

*Its Effects on Papua New Guinea's
Economy, Environment, and People*

*Baptist Missionary School
Pangia*

In Word

In Conversation

In Charity

In Spirit

In Faith

In Purity

1 Timothy 4:12

Be thou an example

*"He who is faithful in what is least is faithful also in much."
Luke 16:10*

Prepared by: **Stan Tiki**
Date of Birth: **2011**
School: **Pangia Baptist Missionary School**
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*"The LORD God took the man and put him in the Garden of Eden
to work it and take care of it."*

Genesis 2:15

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NAME *Stan Tiki*

DOB; *2011*

SCHOOL; *Pangia Baptist Mission School.*

Reason for My Research

I chose to research El Niño because I heard that another El Niño event could happen. I wanted to know what El Niño is, how it happens, and how it can affect Papua New Guinea and the people living here.

I was also interested in learning what we can do to prepare before problems happen. El Niño can affect our water, food gardens, health, and daily lives, so I wanted to understand more about it and find ways people can stay safe.

What Is El Niño?

Before I started my research, I only knew that El Niño means dry season. but now I learned that El Niño is a natural weather event that happens when the surface water in parts of the Pacific Ocean becomes warmer than usual.

El Niño can change weather patterns in many countries. Some places may get very little rain and experience drought, while other places may get too much rain and experience flooding.

Even though the ocean temperature only changes by a small amount, it can still have a big effect on the weather and people's lives.

How El Niño Forms

Normally, trade winds blow across the Pacific Ocean and push warm water towards Australia and Asia. At the same time, cooler water rises near South America.

During El Niño, these winds become weaker, which allows warm water to move back towards the eastern Pacific Ocean. This changes the way heat and air move around the Earth.

Scientists know an El Niño event is happening when the ocean temperature stays about 0.5°C warmer than normal for several months. This small change can cause some places to become very dry while others may receive heavy rain.

What NASA Says

While doing my research, I found out that NASA uses satellites to study the Pacific Ocean. These satellites help scientists watch changes in ocean temperatures and understand when an El Niño event may happen.

I also learned that warmer ocean water can affect sea life because some fish move to cooler areas. This can make fishing harder for some communities in our country.

NASA explains that El Niño can cause different weather problems around the world, including droughts, floods, heatwaves, and bushfires.

How El Niño Can Affect Papua New Guinea

During my research, I found out that El Niño can affect many parts of Papua New Guinea, especially places where people depend on rainfall to grow their food. The Highlands region is one of the areas that can be seriously affected because many families rely on gardens for their daily meals.

In provinces such as Southern Highlands, Enga, Western Highlands, Simbu, Eastern Highlands, and Hela, drought can cause water shortages and make it difficult for people to grow enough food. When the rain does not come for a long time, rivers and water sources can become smaller, and gardens can start to dry out.

In Pangia the weather is wet and dry or mixed so when the drought came the weather went dry and cold creating high pressure which makes rain hard to fall. Some high-altitude areas can also face other problems. In places like Tambul, Nebilyer, and areas close to Mount Giluwe, frost and hail have damaged or destroyed gardens cutting off the supply of food sources. When gardens are affected by drought, frost, or hail, it can increase the risk of food shortages,

hunger, and poor nutrition, especially for young children. This can cause hunger and malnutrition, especially for young children.

Gardens have been growing well in some places, especially near the river where the ground is usually wet all year round and in the swampy areas. People are only scared of frost and hail because it might destroy their gardens affecting crops that people might feed upon .

El Niño does not affect only one place. It can affect many communities across Papua New Guinea by changing weather patterns, reducing food supplies, and making everyday life more difficult.

Areas and Things That Can Be Affected

Different parts of Papua New Guinea can experience different problems during El Niño.

The Highlands we can be affected by drought and frost. Frost can destroy gardens and make it harder for people to grow food. The Southern parts of PNG, including places like Western Province, Gulf Province, Central Province, and the National Capital District, can experience long dry periods. Water can become harder to find, and dry land can increase the chance of bushfires. Other areas, including parts of Momase and the New Guinea Islands, can also experience less rainfall. This can affect gardens, water supplies, and daily life. During droughts, clean water can become difficult to get. If people use dirty water, diseases like diarrhoea and typhoid can increase. I don't know about other places but in Pangia it's getting really dry but there's little rain falling like once every 2-3 weeks. After a long dry period, heavy rain can sometimes leave water standing around homes. This can become a place where mosquitoes breed. Because of this, diseases like malaria and dengue fever can spread more easily. Dry conditions can also create more dust and smoke from bushfires. This can cause breathing problems for some people.

This research helped me understand that El Niño is not only a weather problem. It can affect people's health, food, water, and everyday lives.

What We Can Do

During my research, I found out that El Niño can affect many parts of Papua New Guinea, especially places where people depend on rain to grow their food. The Highlands region is one of the areas that can be seriously affected because many families rely on gardens for their daily meals.

In provinces such as Southern Highlands, Enga, Western Highlands, Simbu, Eastern Highlands, and Hela, drought can cause water shortages and make it difficult for people to grow enough food. I learned that when the rain does not come for a long time, rivers and water sources can become smaller, and gardens can start to dry out.

Some high-altitude areas can also face other problems. In places like Tambul, Nebilyer, and areas close to Mount Giluwe, frost and hail have damaged gardens. This surprised me because I usually think about drought only as a lack of rain, but I learned that cold weather events can also affect food production.

From what I have seen and heard in my area, dry conditions can make life harder for people because they have to look for water and protect their gardens. In some places in Pangia in the swampy areas food is growing really well but other parts of the mountain areas are getting dry but thankfully God sent rain to us. But overall it's all growing luxuriously. The only problem is with drinking water because of lack of rain. Tanks got dry and people had to go to the river to wash clothes and fetch drinking water at small creeks that came out of rocks.

El Niño does not affect only one place. It can affect many communities across Papua New Guinea by changing weather patterns, reducing food supplies, and making everyday life more difficult. Something I learned during my research is that El Niño can also affect people's health. Before I started researching, I did not know that changes in weather could make some diseases spread more

easily. From this research, I learned that El Niño is not only a weather problem. It can affect people's health, water, food, gardens, and everyday lives.

CONCLUSION

In conclusion, El Niño is a natural weather event that can have a big impact on Papua New Guinea. It can cause drought, water shortages, frost, crop damage, food shortages, and some diseases.

Even though not every part of the country is affected in the same way, everyone should understand what El Niño is and be prepared. Learning about El Niño and taking simple steps early can help protect our families, our communities, and our country.

Biblical Point of View

The Bible teaches that God created the earth and everything in it. He gave people the responsibility to look after His creation. In **Genesis 2:15**, God told Adam to work in the Garden of Eden and take care of it. This reminds us to care for the environment and use natural resources wisely.

The Bible also teaches us to prepare for the future. **Proverbs 21:5** says that careful planning leads to success. This encourages us to prepare for events like El Niño by saving water, protecting our food gardens, and helping others.

When people are affected by drought or food shortages, Christians should show kindness and help those in need. **Galatians 6:2** reminds us to care for one another and share each other's burdens.

As Christians, we can pray for God's help while also doing our part by preparing wisely and looking after the resources He has given us.

Sources

NASA

National Oceanic and Atmospheric Administration (NOAA)

Papua New Guinea National Weather Service

Papua New Guinea National Department of Health

World Health Organization (WHO)

thankyou